

Louis Messonnier

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

May 2028

Bachelor of Science in Mechanical Engineering, Minor in Materials Science and Engineering

GPA: 4.00

EXPERIENCE

HyTech Racing Georgia Tech, Atlanta, GA

September 2024-Present

Powertrain Engineer

- Performed continuous assembly, disassembly, and troubleshooting of powertrain components including gearbox, uprights, and cooling system throughout each year
- Set up, maintained, and ran an Arduino-based cooling test rig across testing cycles, logging temperature, pressure, and flow rate to evaluate cooling system performance
- Designed, analyzed, and prototyped powertrain components using tools including CATIA, nTop, FEA, and 3D printing, integrating parts within Formula SAE packaging and rule constraints
- Evaluated powertrain component selection using datasheets and performance requirements to ensure system compatibility

Duke University Thomas Lord Department of Mechanical Engineering & Materials Science, Durham, NC

Summer 2025

Undergraduate Research Assistant

- Planned and carried out a self-guided study on the mechanical behavior of anisotropic gyroid unit cells under tensile and compressive loading following an extensive literature review on TPMS lattices, porous biomaterial mechanics, bone-implant interface behavior, and 3D-printed surface topography
- Produced and tested 128 SLA-printed samples across two load types, two stretch directions, four stretch ratios, and two base unit cell sizes
- Processed and analyzed mechanical test data using MATLAB and Excel, including stress-strain computation, property calculation, and graphing
- Identified trend in anisotropic tensile mechanical behavior: stretching unit cells parallel to loading increased Young's modulus while holding strength steady, whereas perpendicular stretch reduced both
- Built an Ansys tensile simulation of gyroid samples that qualitatively matched experimental fracture behavior, gaining experience in meshing, boundary conditions, and material modeling

PROJECTS

HyTech Racing – Topology Optimized Upright

- Built an nTop topology optimization workflow for the vehicle's front upright for additive manufacturing in Ti-6Al-4V including preserved interfaces and boundary conditions with loads and restraints for combined load cases
- Validated nTop model against the existing FEA, matching von Mises stress and displacement before optimizing

Personal Project – Fully Custom Magnetic Switch Keypad

- Generated PCB schematic and layout for AT32F405-based keypad with three Hall-effect magnetic switches, six mechanical switches and one rotary encoder based on hardware requirement documentation of microcontroller
- Included linear Hall sensors under each analog switch to measure continuous key travel, enabling adjustable actuation points and 8 kHz USB polling through the microcontroller's high-speed USB
- Designed and iterated a 3D-printed case in SolidWorks, then ordered PCBs and soldered, assembled, and tested the finished keypad
- Altered existing open source Hall-effect keyboard firmware to add support for mechanical switches and rotary encoder

SKILLS

Technical:

- CAD:** SolidWorks, CATIA, nTop, KiCad
- Simulation:** Ansys Mechanical, nTop
- Programming:** MATLAB, Arduino
- Fabrication:** SLA/FDM 3D printing, mechanical testing systems

Relevant Coursework: Statics, Dynamics, Circuits, Electronics, Engineering Computing, Materials Science, Engineering Graphics, Thermodynamics, Fluid Dynamics, Deformable Bodies

Commented [LM1]: Add anything else I did/do: gyroid infill, suspension arm geometry from matlab calcs, ntop automate, enveloped model

Commented [LM2]: Add how validation worked (how much match) "Validated the nTop static model against the team's CATIA/SIMULIA FEA, matching peak von Mises stress within 1% before optimizing"

Commented [LM3]: Add "Reduced upright mass by [X]% while keeping peak stress under [Y] MPa across all load cases in verification solves on the optimized geometry"